University of British Columbia, Vancouver
Department of Computer Science

CPSC 304 Project Cover Page

Milestone #: 1

Date: January 23rd, 2026

Group Number: 91

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

2. A brief project description answering these questions:
a. What is the domain of the application? Describe it.
The domain of an application refers to the area of knowledge your application resides in. For example, if I am making an application for a hospital, the domain would be something like healthcare/patient management/logistics (it would depend on what the application is trying to do). Ensure that all members of the team are familiar with the domain and data that will be modelled. If you do not understand the domain/data, you will have an exceptionally hard time meaningfully contributing to the project which will lead to lots of project issues down the road.

Our Open-Source Dependency Risk & License Compliance Database aims to solve the problem of keeping track of legal obligations, security risks, and policy requirements when it comes to hundreds, if not thousands of open-source libraries (dependencies in this case), that are referenced in modern day software projects. Furthermore, offering potential solutions along the way as a form of remediation. In this case, the domain of this project would be logistics, version control, legality and software management.

b. What aspects of the domain are modeled by the database? In answering this question, you will want to talk about what your project is trying to address and how it fits within the domain. It is likely that in the process of answering these questions you will bring up examples of a real-life situation that the application could be applied to.

This system would act as a central registry and monitoring platform that tracks what open-source code is currently being used in your projects, along with legal compliance, i.e. what licenses apply to each dependency,

Aspects of the domain modeled by the database:
- Intellectual Property (IP) / Licenses, when using open source code / libraries
- Keeping track of what technologies are being used (for both your benefit, and for legal reasons)
- Plagiarism concerns (Academic use / Self-gain)
- Solutions workflow in the case of a security/license risk

Real-life situations that the application could be applied to:
- CS departments could cross-reference a project and its open source dependencies, to detect plagiarism
- Legal teams can protect their IP by cross-referencing a potential infringement case
- As a developer, the database could highlight the misuse of licenses, etc.
- A critical vulnerability is discovered in the open-source code of one the dependencies referenced in a project(s)
    - System scans and finds all projects using the open-source code and generates a notification.
    - Log4Shell – https://en.wikipedia.org/wiki/Log4Shell

3. An ER diagram for the database that your application will use. It is OK to hand-draw it but if it is illegible or messy or confusing, marks will be taken off. You can use software to draw your diagram (e.g., draw.io, GoogleDraw, Microsoft Visio, Powerpoint, Gliffy, etc.) The result should be a legible PDF or PNG document. Note that your ER diagram must use the conventions from the textbook and the lectures. For example, do not use crow's feet notation or notation from other textbooks).
a. Please limit your diagram to a letter size page (8.5 x 11 inches). If you require additional space, talk to your project mentor beforehand as this might mean that your project is a bit more complicated than what we expect.

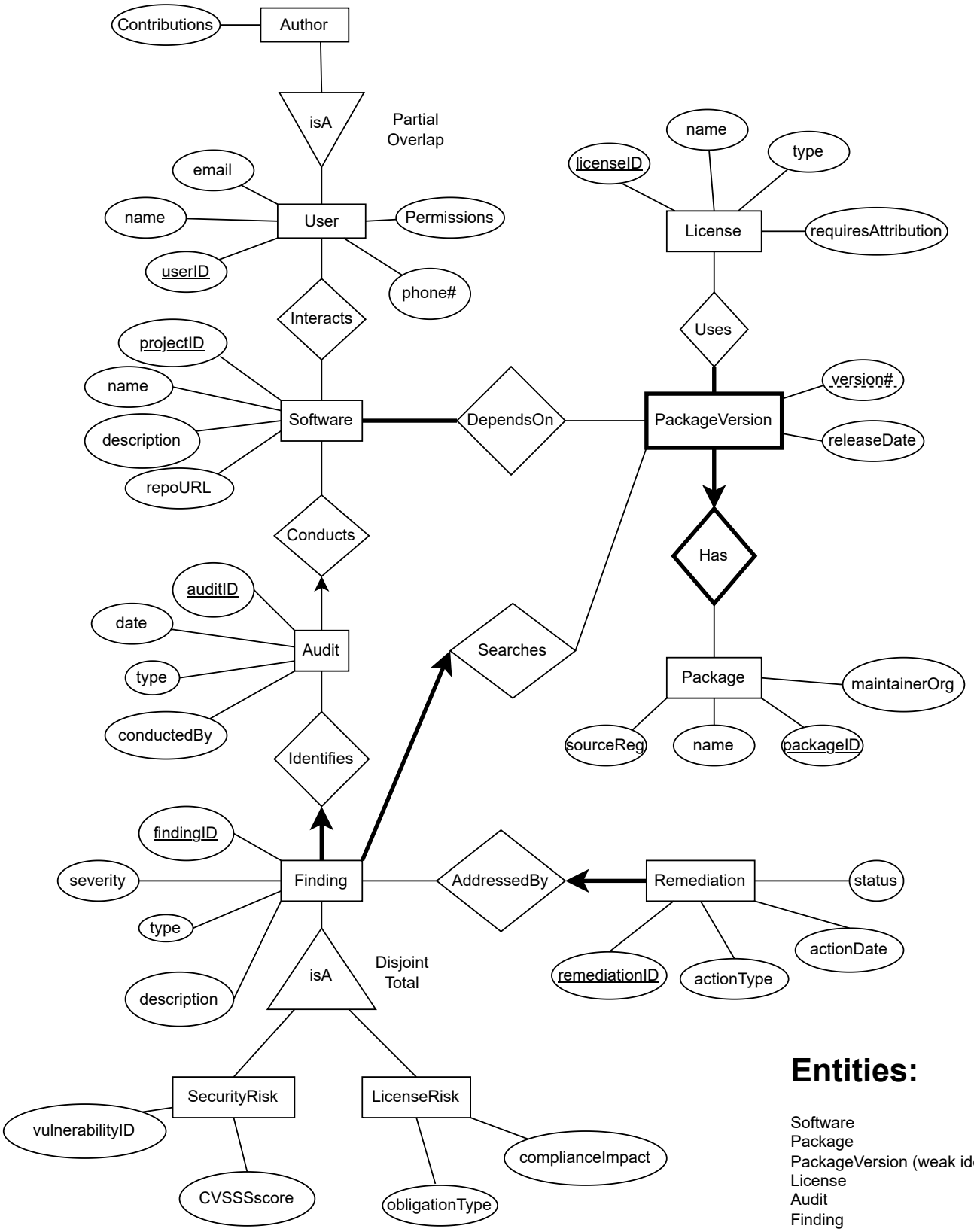
ER Diagram

5. Other comments, as appropriate, to explain your project.

- Finding: Some risk associated with licensing/package version
- Remediation: Some suggested fix to a finding

6. An explicit acknowledgment about your use of AI tools in this assignment. Specifically, we are looking for a clear yes/no about whether you have used one or more AI tools. If yes, we want to know which tool(s) you have used and we want a PDF of the full conversation.

- No AI tools were used in the creation of this milestone.



Entities:

Software
Package
PackageVersion (weak identity)
License
Audit
Finding
Remediation
User